GitHub Agentic Workflows

PUBLIC PREVIEW

MONA LISA
Mascot
GitHub

MADE WITH ♡ BY
GitHub Next, Microsoft Research,
Azure Core Upstream & GitHub Actions

Repos that clean themselves

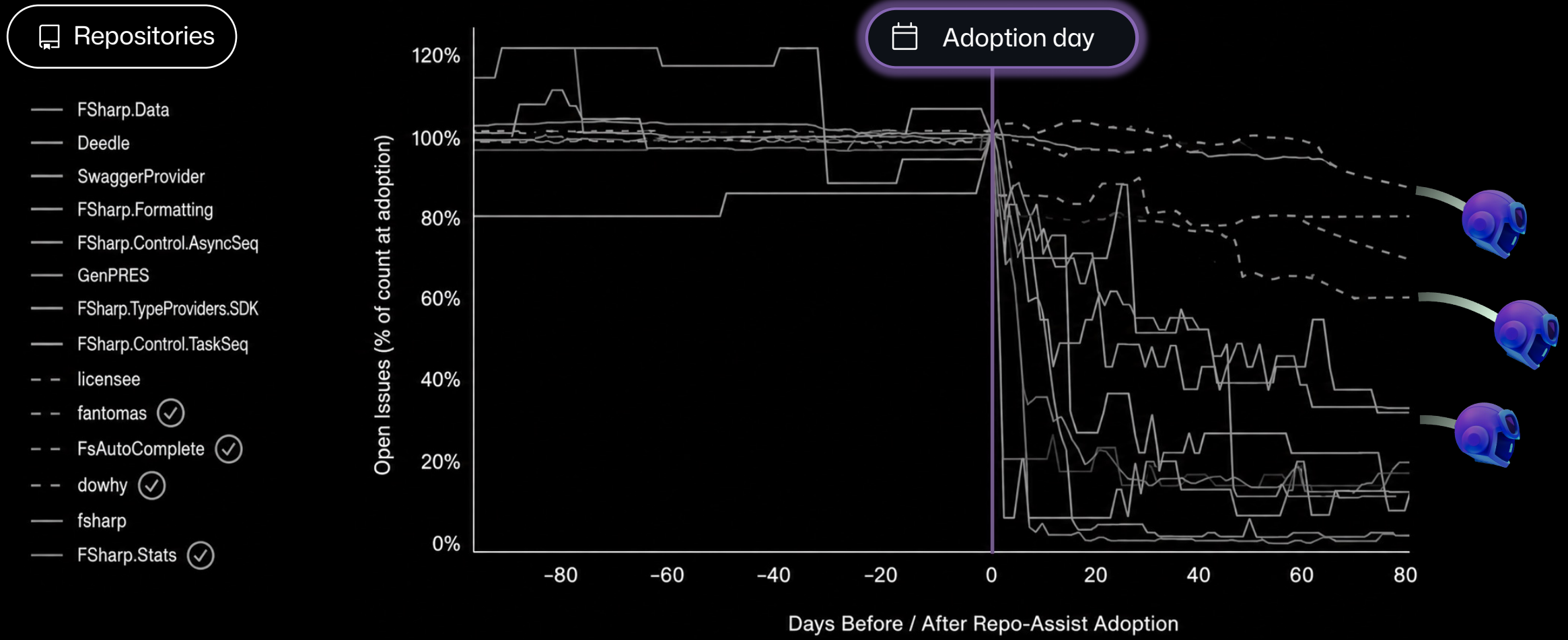
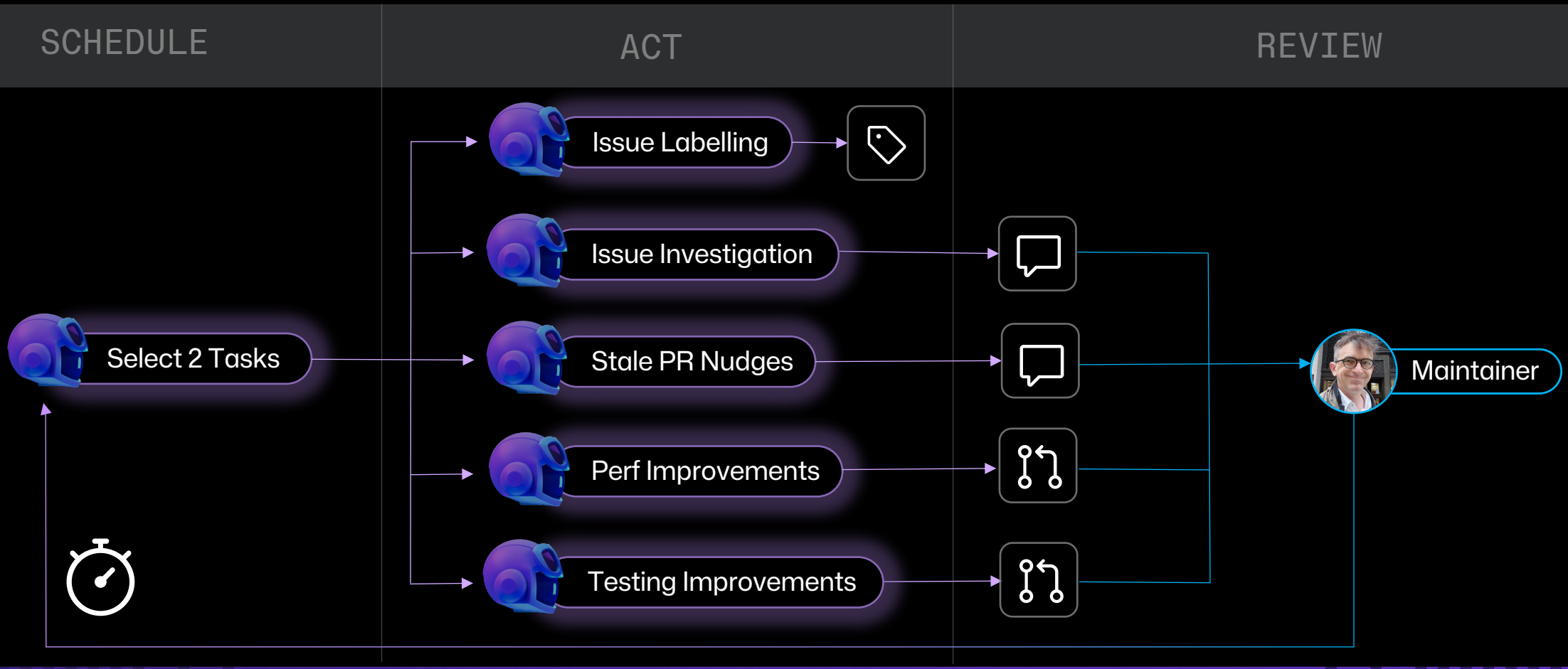
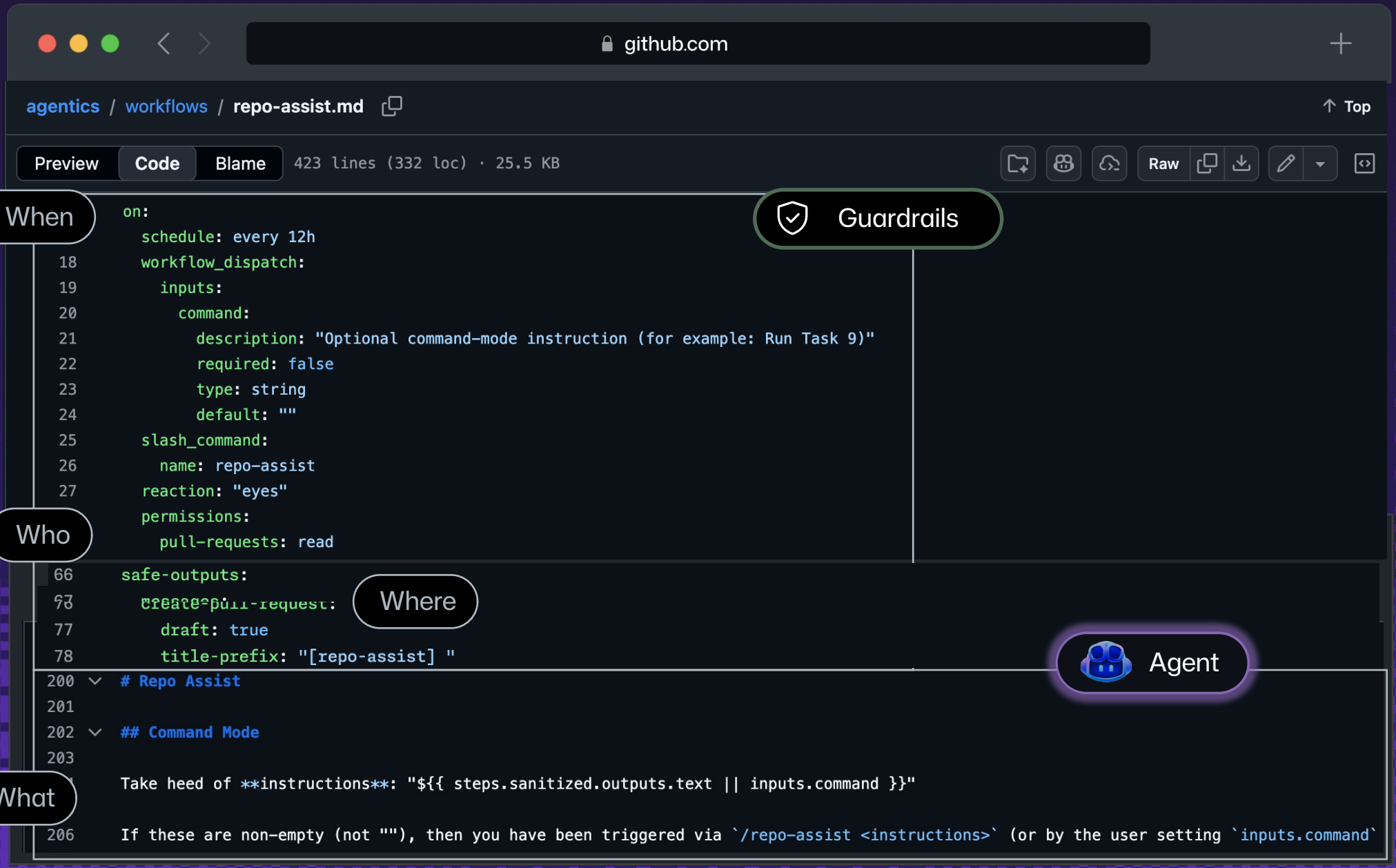


Figure: Burning down open issues in F# repositories with “Repo Assist” workflow.

Repo Assist

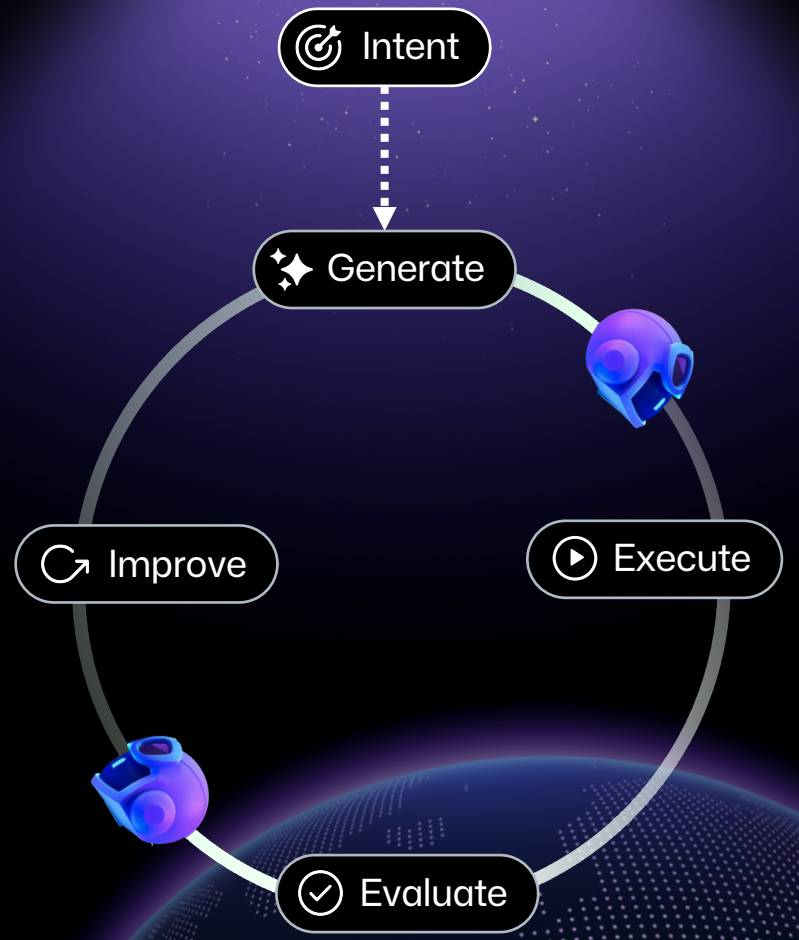
Auto Maintenance Loop




github.com
agentics / workflows / repo-assist.md
Top
Preview
Code
Blame
423 lines (332 loc) · 25.5 KB
Raw
When
Guardrails
Who
Where
Agent
What
on:
schedule: every 12h
workflow_dispatch:
inputs:
command:
description: "Optional command-mode instruction (for example: Run Task 9)"
required: false
type: string
default: ""
slash_command:
name: repo-assist
reaction: "eyes"
permissions:
pull-requests: read
safe-outputs:
create-pull-request:
draft: true
title-prefix: "[repo-assist] "
Repo Assist
Command Mode
Take heed of **instructions**: "${{ steps.sanitized.outputs.text || inputs.command }}"
If these are non-empty (not ""), then you have been triggered via `/repo-assist <instructions>` (or by the user setting `inputs.command`

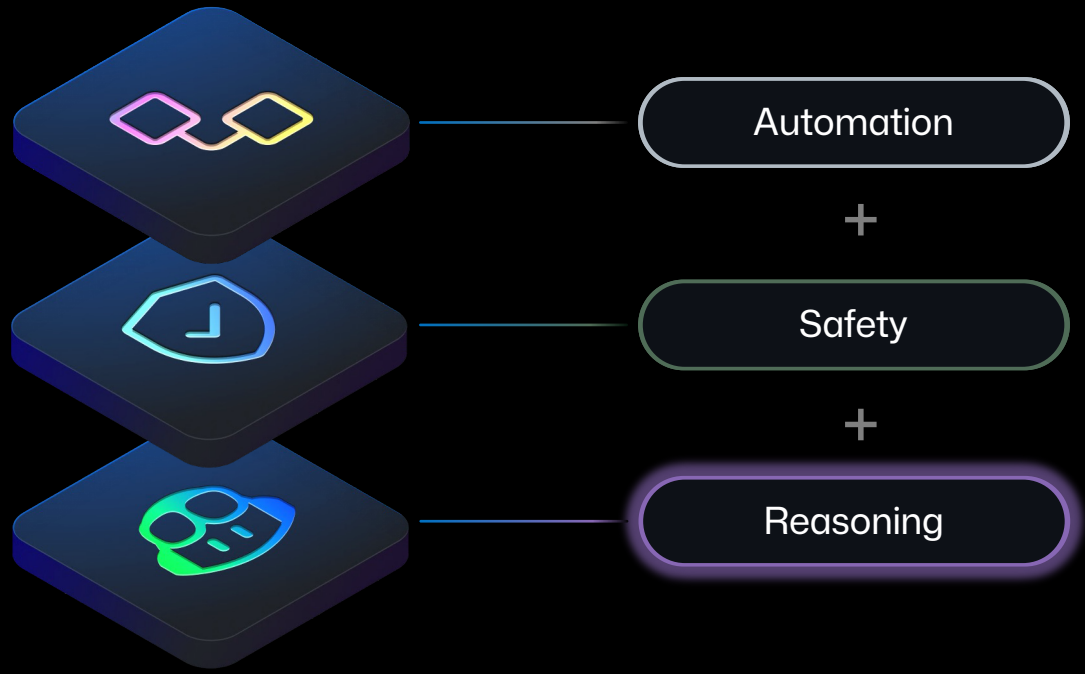
Human Agent Processes

- **Agents accelerate the entire SDLC.**
  Every step of our software process can be “agentified”.
- **Automation is the true accelerator.**
  Completion: 1.4x, Dev solo: 10x, Automation: 100-1000x.
- **Scale with processes not swarms.**
  Orchestration not chaos.



GitHub Agentic Workflows

Human Agent Processes powered GitHub.



Workflows executed via GitHub Actions written in natural language.

Isolated sandboxes with scoped permissions and network control.

Agents interpret intent & operate on context across models. Copilot CLI · Claude · Codex · Gemini

Orchestrated Automation
through Human Agent Processes

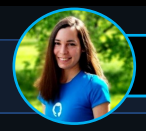
issue-clarifier.md

.github > workflows > issue-clarifier.md > # Issue Clarifier

```
1   ---
2   on:                          # Trigger: when to run
3     issues:
4       types: [opened]
5
6   permissions: read-all        # Security: read-only by default
7
8   safe-outputs:                # Allowed write operations
9     add-comment:
10  ---
11  # Issue Clarifier
12
13  Analyze the current issue and ask for additional details
14  if the issue is unclear.
15
```

Sandbox

Agent



Developer

mnkiefer opened 24 minutes ago

Please add better setup instructions.

Right now it's not clear how to get started, what prerequisites are needed, or what the expected workflow is.

Create sub-issue

github-actions bot 20 minutes ago – with GitHub Actions

Thanks for raising this! The request for better setup instructions is clear, but to write something accurate and useful, a few more details would help:

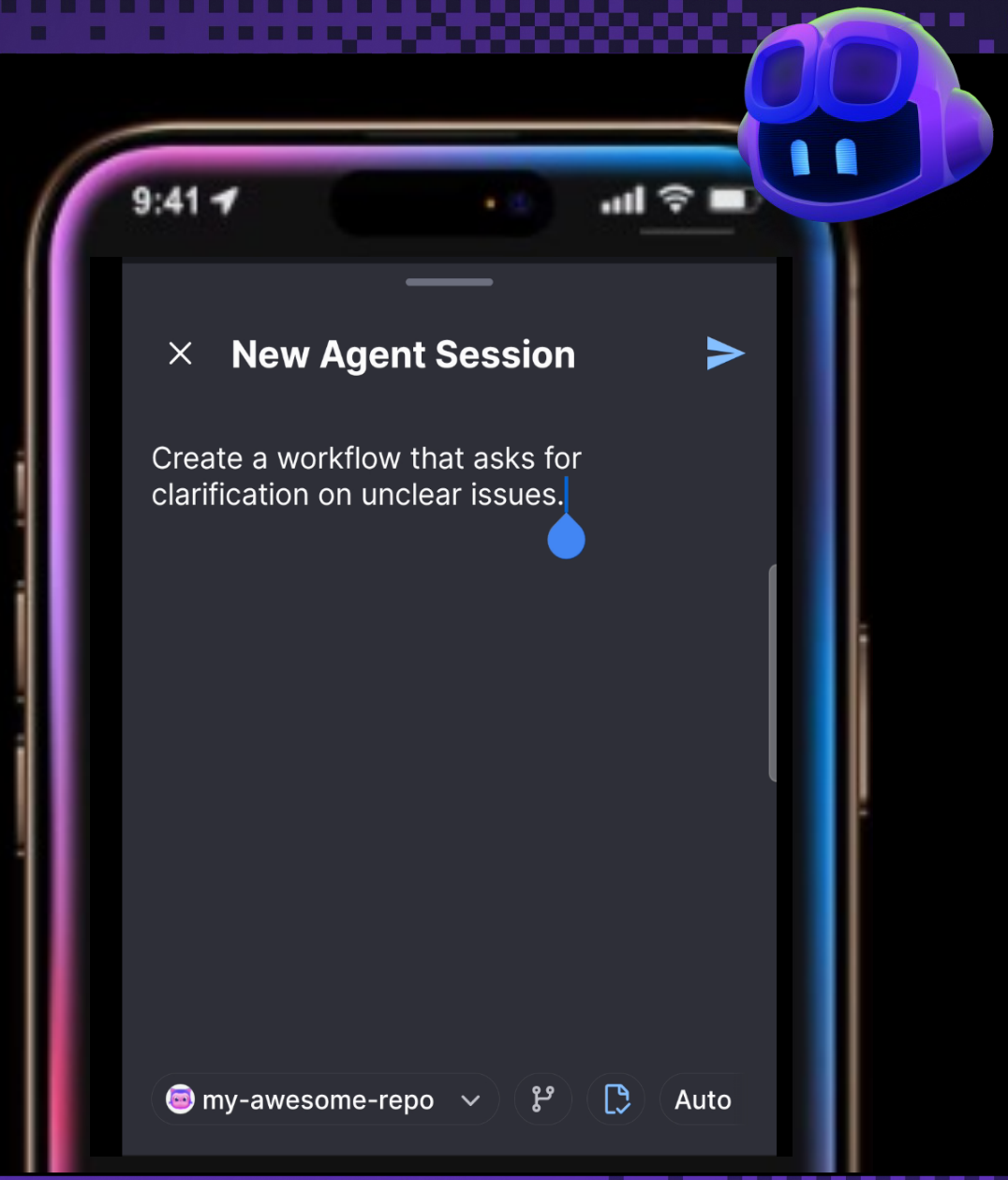
1. **What does this project do?** A brief description of the project's purpose would help frame the setup instructions appropriately.

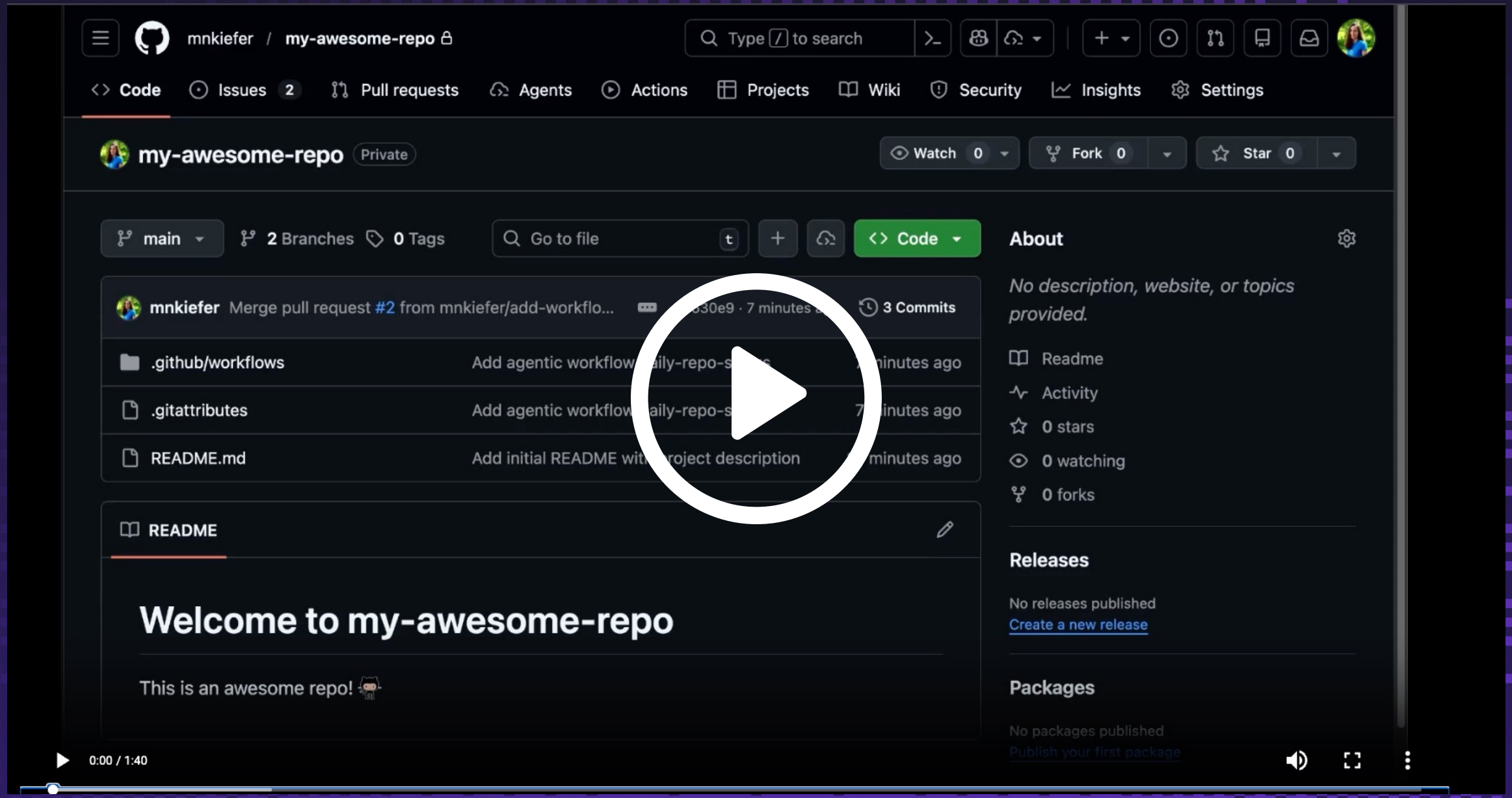
Spec Driven Automation
through Agentic Authoring

Write agents with the help of agents, iteratively.

- ✓ Configure your repository
  for GitHub Agentic Workflows
- ✓ Create new workflows
- ✓ Debug existing workflows
- ✓ Optimize Token Usage

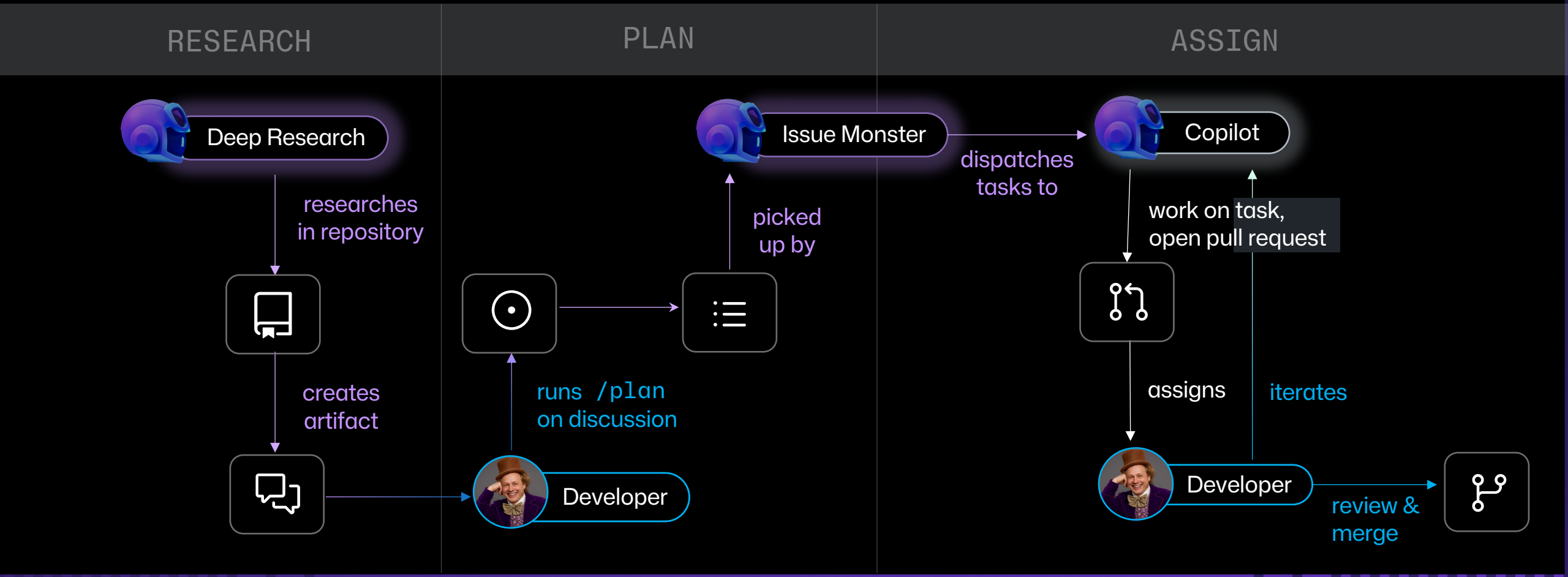
Spec Driven Development




mnkiefer / my-awesome-repo
Type / to search
Code
Issues 2
Pull requests
Agents
Actions
Projects
Wiki
Security
Insights
Settings
my-awesome-repo
Private
Watch 0
Fork 0
Star 0
main
2 Branches
0 Tags
Go to file
t
Code
mnkiefer Merge pull request #2 from mnkiefer/add-workflo...
3 Commits
.github/workflows
.gitattributes
README.md
README
Welcome to my-awesome-repo
This is an awesome repo!
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
0:00 / 1:40

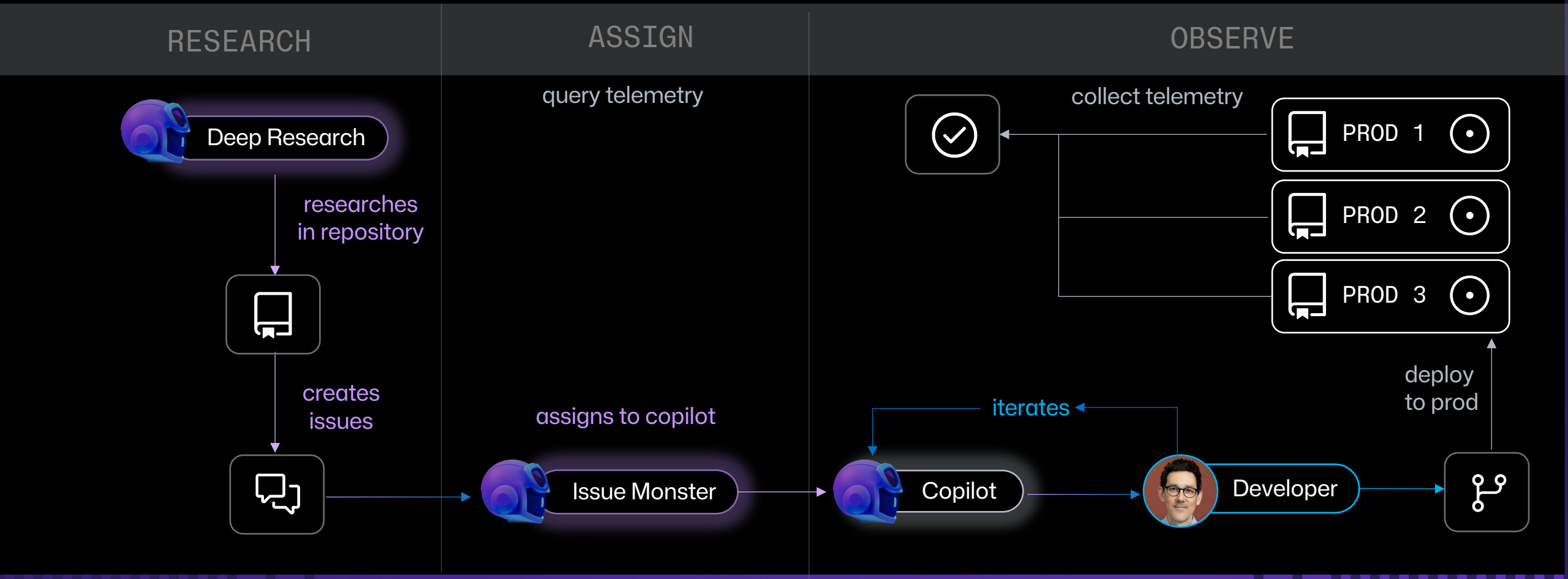
Research – Plan – Assign

Daily Test Improver, Daily Perf Improver, ...



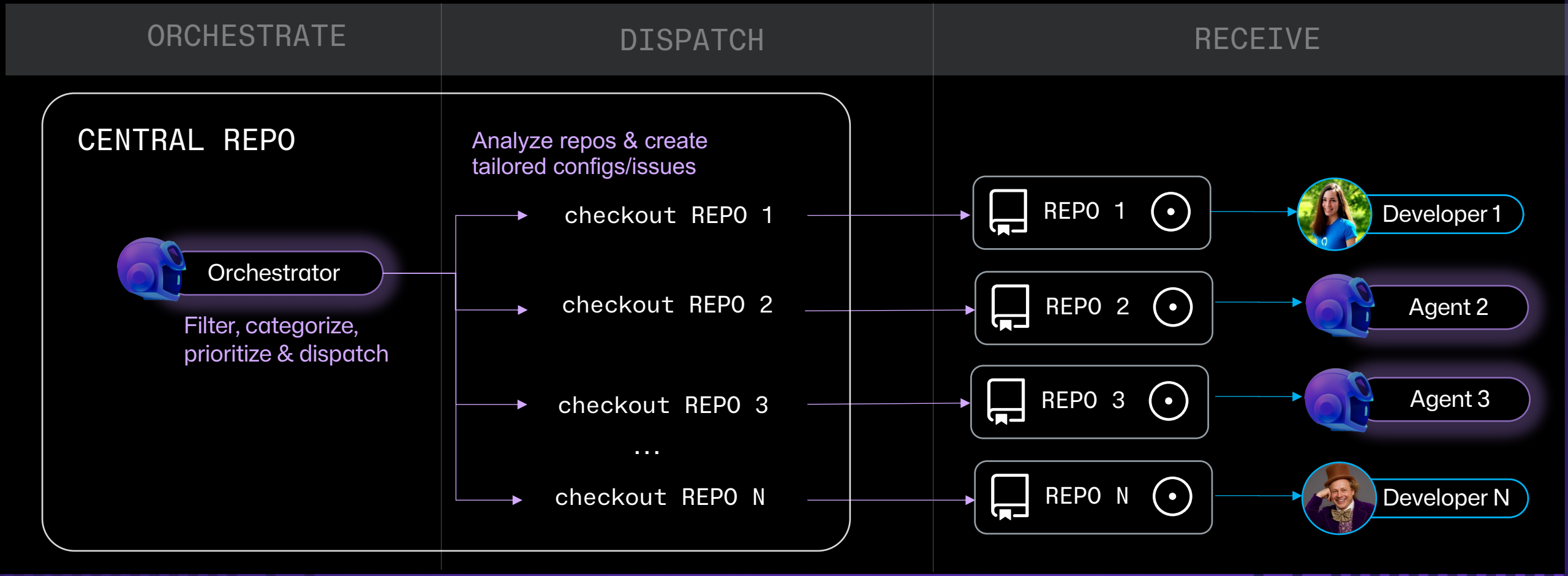
Research – Assign – Observe

Agentic Feedback Loop

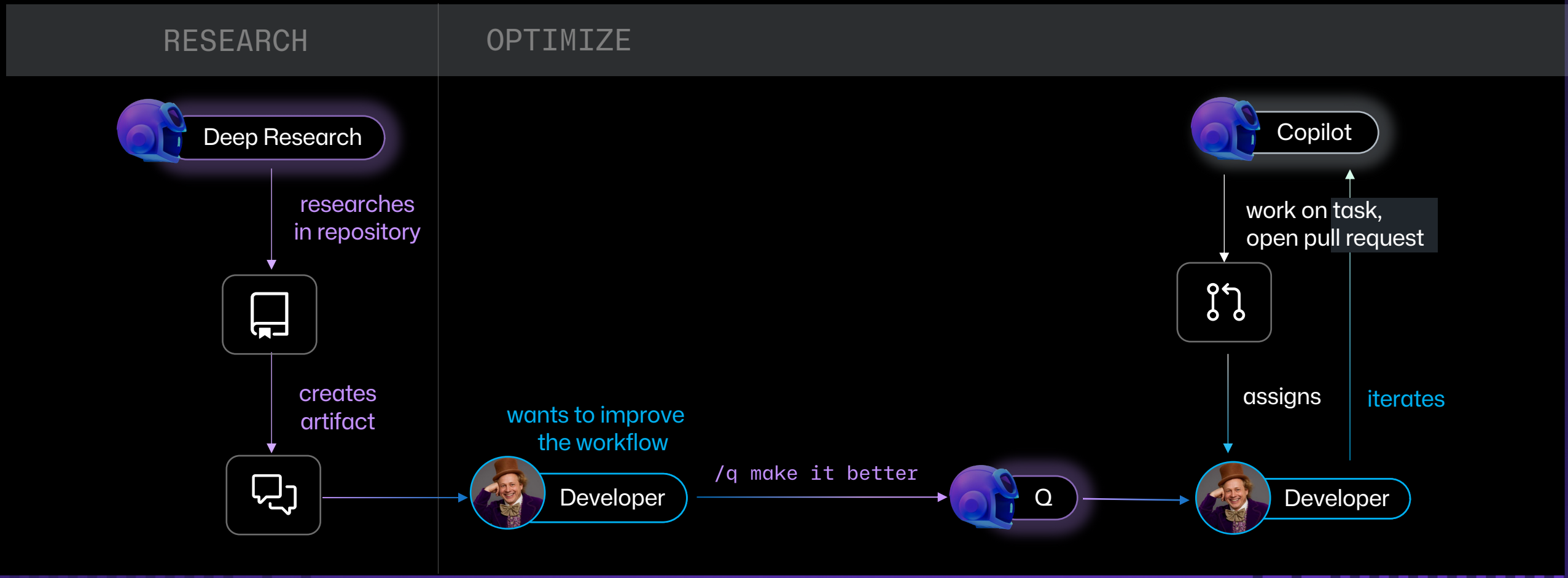


Central Repo Ops

Rolling out large scale operations



The Agent Improver /q



At scale, agentic CI creates a new problem

The hidden token tax

- **Costs accumulate quietly**
  Agentic CI jobs runs automatically.
- **Small inefficiencies compound**
  A few wasted calls per run become expensive.
- **Raw token counts can mislead**
  Model choice, output tokens, workload size, etc.

Cost Management

to guard, measure, optimize

daily-pr-summary.md

.github > workflows > daily-pr-summary.md > # Daily PR Summary

```
1  ---
2  on: daily
3  tools:
4    github: pull_requests
5  safe-outputs:
6    create-issue:
7  ---
8  # Daily PR Summary
9
10 Create an issue with the executive summary of the activity
11 in the repo yesterday.
```

daily-pr-summary.md

daily-pr-summary.md > # Daily PR Summary > ## models: haiku

```
1  ---
2  on: weekly
3  tools:
4    github: pull_requests
5  safe-outputs:
6    create-issue:
7  steps: # deterministic compute
8    - run: gh pr list -after -1d > prs.json
9  max-ai-credits: 500
10 max-daily-ai-credits: 1000
11 ---
12 # Daily PR Summary
13
14 (1) Summarize each PR in prs.json with the `pr-summary` subagent
15 (2) Create executive summary from PR summaries into an issue
16
17 ## agent: `pr-summary`
18 ---
19 models: haiku
20 ---
21 Create an executive summary of the PR.
```

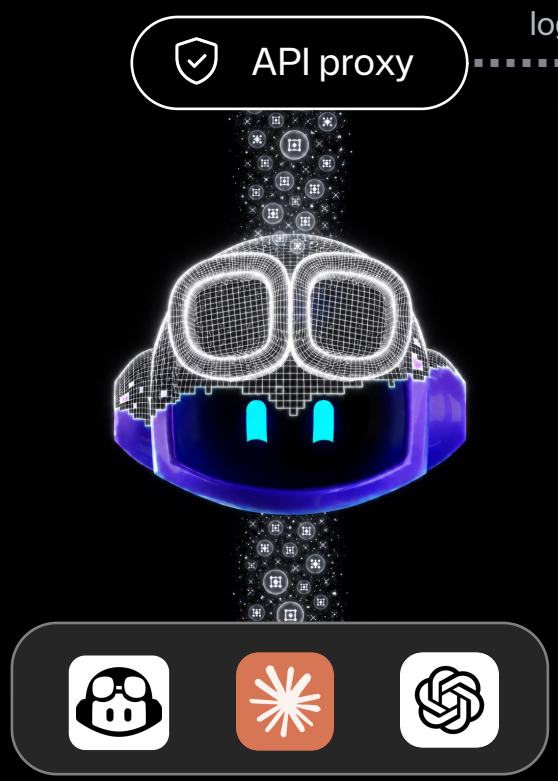
Scheduling

Deterministic

Cost

Sub-Agent

Make every token traceable



AI Credits = $ 0.01 USD

#	Alias	Input	Output	Cache Read	Cache Write	ΔAI Credits	AI Credits	Duration
1	opus48	8,070	20	0	0	4.08	4.08	6.7s
2	opus48	1	564	0	9,361	7.26	11.3	6.7s
3	opus48	2	584	9,361	1,934	3.14	14.5	7.5s
	...					...		
29	opus48	2	621	39,123	790	4	95.4	8.8s
30	opus48	2	318	39,913	752	3.26	98.7	4.5s
Total		**8,127**	**11,630**	**803,564**	**40,665**		**94.7**	**2m58s**

Table: Token usage for a “Smoke Claude” workflow run.

Make every token count

Quick wins

Remove obvious waste

- ❑ Prune unused MCP calls
- ❑ Stop repeated tool calls
- ❑ Fix fallback loops

Structural fixes

Move work out of the agent

- ❑ Pre-fetch diffs, files, metadata
- ❑ Add gates to skip irrelevant runs
- ❑ Use CLI proxies for deterministic reads

Strategic leverage

Optimize the system

- ❑ Right-size models by task
- ❑ Split monoliths into subagents
- ❑ Run A/B experiments
- ❑ Tune scheduling

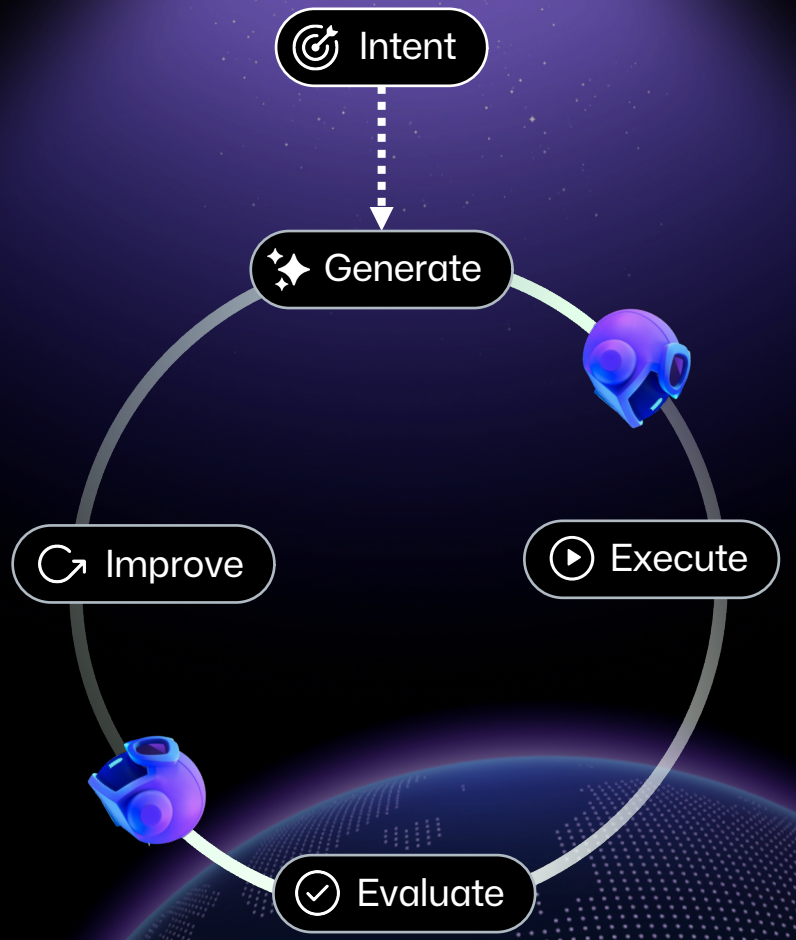
Make optimization continuous

Install the GitHub Agentic Workflows CLI:

```
gh extensions install github/gh-aw
```

Add our automatic token audit & optimizers to your central org repository:

```
gh aw add githubnext/agentic-ops
```



Agents are power tools

Agentic workflows are machine tools

Human Agent Processes are factories

Patterns

See our research for more useful design patterns.

[gh.io/gh-aw](gh.io/gh-aw)

Welcome to Peli's Agent Factory

Jan 12, 2026

Don Syme Peli de Halleux Mara Kiefer

Welcome, welcome, WELCOME to Peli's Agent Factory!

Imagine a software repository where AI agents work alongside your team - not replacing developers, but handling the repetitive, time-consuming tasks that slow down collaboration and forward progress.

Peli's Agent Factory is our exploration of what happens when you take the design philosophy of **"let's create a new automated agentic workflow for that"** as the answer to almost every opportunity that arises! What happens when you **max out on automated agentic workflows** - when you make and use dozens of specialized, automated AI agentic workflows and use them in practice.